Skills Summary

Whole-Number Division: Up to Four Digits by One-Digit Divisors

Use this reference while completing your worksheet or when studying.

Skill 1 — The long-division cycle

The cycle: divide → multiply → subtract → bring down.

Start at the biggest place value the divisor will fit into, then turn the cycle once more for every digit still waiting to come down.

Words to keep straight: in $852 \div 4$, the *dividend* is the number being split, the *divisor* is the size of each group, and the *quotient* is the answer.

Always check: quotient $\times$ divisor = dividend.

Example: $852 \div 4$

Step 1. We are splitting 852 into groups of 4, and 4 does fit into the hundreds digit, so run the cycle from the hundreds place downward.

Step 2. Hundreds: $8 \div 4 = 2$, nothing left over. Tens: $5 \div 4 = 1$ with 1 left over; bring the 2 down beside that leftover to make 12.

Step 3. Ones: $12 \div 4 = 3$. Read the quotient across the top: **213**, and the check works: $213 \times 4 = 852$.

$$\begin{array}{r} 213 \\ 4\overline{)\,852} \end{array}$$

Try it: $675 \div 5$

check: **135**

Skill 2 — Four-digit dividends and zeros

Rule:

A fourth digit just means one more turn of the same cycle. If the number you are dividing at some step is *smaller* than the divisor, that place gets a **0** in the quotient — write it, then bring down the next digit. Every place you divide into earns a digit.

Example: $3{,}612 \div 6$

Step 1. Six will not fit into 3 thousands, so the first place we can actually divide is the hundreds — which already tells us the quotient has three digits.

Step 2. $36 \div 6 = 6$ hundreds, nothing left. The next digit down is 1, which is too small for 6, so the tens digit of the quotient is 0.

Step 3. Bring down the 2 to make 12: $12 \div 6 = 2$ ones, giving **602**. Check: $602 \times 6 = 3{,}612$.

Try it: $5{,}145 \div 5$

check: **1029**

(!) Watch out: dropping that 0 turns 602 into 62 — and 62×6 is only 372, nowhere near 3,612. The multiplication check catches this mistake every time.

Skill 3 — Remainders, and what they mean

Rule:

dividend = divisor × quotient + remainder, and the remainder is always *smaller* than the divisor.

Then ask what the leftover means in the story: **ignore** it (how many *full* boxes), **round up** (how many buses are needed), or **report** it (how many are left over).

Example: $437 \div 8$
Step 1. We want how many 8s fit inside 437; 8 will not go in evenly, so divide as far as possible and then name what is left.
Step 2. $43 \div 8 = 5$ tens with 3 left over, since $8 \times 5 = 40$. Bring down the 7 to make 37: $37 \div 8 = 4$ ones with 5 left over.
Step 3. That 5 is smaller than 8, so it stays as the remainder: **54** R **5**. Check: $8 \times 54 + 5 = 437$.

Try it: $293 \div 7$

check: **41** R **6**

Skill 4 — Estimate first, then check

Rule:

Trade the dividend for a *compatible number* — a nearby number the divisor goes into evenly — and divide that instead. The estimate tells you how big the real quotient should be, so a digit written in the wrong place shows up immediately. Afterwards, multiply the quotient back by the divisor to check.

Example: About how much is $3{,}489 \div 6$?
Step 1. We only need a close answer here, so swap the dividend for a friendly multiple of the divisor instead of dividing exactly.
Step 2. 3,600 sits close by and 6 divides it evenly, so estimate with $3{,}600 \div 6 = 600$.
Step 3. The estimate is **600**, so the exact quotient has to land near 600 — an answer like 58 or 5,800 would have to be a place-value slip.

Try it: Estimate $2{,}918 \div 7$ with a compatible number.

check: **400**